MAIC - Introduction to Python and Robotics - Smart CuteBot - Examples

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1. Basic Movements

```python
def on_button_pressed_a():

    # Move forwards (speed of left wheel: 0-100, speed of right wheel: 0-100).
    cuteBot.motors(20, 20)


def on_button_pressed_b():

    # Stop the car.
    cuteBot.stopcar()


# When the A button is pressed, run the 'on_button_pressed_a' function.
input.on_button_pressed(Button.A, on_button_pressed_a)

# When the B button is pressed, run the 'on_button_pressed_b' function.
input.on_button_pressed(Button.B, on_button_pressed_b)
```

2. Light Source Detection – Light Sensor

```python
def on_forever():

    # If the light level is greater than a certain number.
    # 0 = pitch black <-> 255 = bright white.
    if input.light_level() > 50:

        # Move forwards.
        cuteBot.motors(20, 20)

    else:

        # Otherwise stop the car.
        cuteBot.stopcar()


# Keep running the 'on_forever' function in the background.
basic.forever(on_forever)
```

3. Object Detection and Avoidance – Ultrasonic Sensor

```
# Global variables that can be accessed by any function.
sonar = 0
cuteBot.motors(20, 20)


def on_forever():
    global sonar

    # Get the distance to the closest object in cm using the ultrasonic sensor.
    sonar = cuteBot.ultrasonic(cuteBot.SonarUnit.CENTIMETERS)

    # If the distance to the closest object is less than a certain distance
    # e.g. 15cm, then stop the car.
    if sonar < 15 and sonar > 1:
        cuteBot.stopcar()
        basic.pause(200)

    else:

        # Otherwise move forwards.
        cuteBot.motors(20, 20)


# Keep running the 'on_forever' function in the background.
basic.forever(on_forever)
```

4. Follow a Route – Line Tracking Sensors

```
def on_forever():

    # If the left line-tracking sensor no longer detects a black line.
    # But the right line-tracking sensor continues to detect a black line.
    # This means that the track is turning right.
    if cuteBot.tracking(cuteBot.TrackingState.L_UNLINE_R_LINE):

        # Turn right by controlling the speeds of the individual wheels.
        cuteBot.motors(20, 0)

    # If the left line-tracking sensor continues to detect a black line.
    # But the right line-tracking sensor no longer detects a black line.
    # This means that the track is turning left.
    if cuteBot.tracking(cuteBot.TrackingState.L_LINE_R_UNLINE):

        # Turn left by controlling the speeds of the individual wheels.
        cuteBot.motors(0, 20)

    # If the left line-tracking sensor continues to detect a black line.
    # And the right line-tracking sensor also continues to detect a black line.
    # That means the track is continuing in a straight line.
    if cuteBot.tracking(cuteBot.TrackingState.L_R_LINE):

        # Move forwards.
        cuteBot.motors(20, 20)


# Keep running the 'on_forever' function in the background.
basic.forever(on_forever)
```